MIL-S-19500/128A
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SUPERSEDING
MIL-S-19500/128 (SigC)
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MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, MIXER

TYPES JAN-1N26B, JAN-1N26BR, JAN-1N26BM, AND JAN-1N26BMR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. *This specification covers the detail requirements for silicon semiconductor diodes, types:* 1N26B (forward polarity); 1N26BR (reverse polarity); 1N26BM (matched forward pair); and 1N26BMR (matched forward and reverse), for use as a mixer (first detector) in a K-band receiver.

1.2 Ratings.

	Type 1/	Z(IF)	L_c	NF_o	VSWR
		ohms	db	db	ratio
Minimum	1N26B	400	---	---	---
Maximum	1N26B	600	7.5	11.0	1.5

1/ Ratings for the 1N26B are applicable to the types having additional suffixes.

OPERATING AMBIENT TEMPERATURE: -65° to +150°C
STORAGE TEMPERATURE: -65° to +150°C

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

Military

MIL-STD-750 - Test Methods for Semiconductor Devices.

DRAWINGS

Armed Services Electro-Standards Agency

104-JAN - Burn Out Test Adapter for Crystal Diode Type 1N26.
107-JAN - Mixer, Test for Crystal Diode 1N26.
108-JAN - Burn Out Tester for Crystal Rectifier 1N21B, 1N23B, 1N23C, 1N23CR, 1N26, 1N28, 1N53 and 1N78.

FSC 5960

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Electronics Command

SCL-5679/1 - Crystal Mixer Sage 221 may be used as an alternate for Drawing 107-JAN

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements for semiconductor diodes shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations and symbols. The abbreviations and symbols used herein are defined in MIL-S-19500, and as follows:

$Z(IF)$ ---------- Intermediate frequency (IF) impedance
R_g ---------- Input resistance of the generator
e_o ---------- Open-circuit voltage of the pulse generator

3.3 Design and construction. The semiconductor diodes shall be of the design, construction, and physical dimensions specified in figure 1.

3.3.1 Plating. The diode shall be plated as specified in figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I and II.

3.5 Marking. The marking shall be placed on each device in accordance with MIL-S-19500.

3.5.1 The manufacturer, at his option may omit the following markings:

(a) Country of origin.
(b) Manufacturer's identification.

3.5.2 Diodes meeting the matching requirements (see 4.5.3 and 4.5.4) of this specification will be packaged with a statement to that effect.

3.6 Burnout by single pulse. At the end of all manufacturing processes and prior to selecting samples for testing, all diodes shall be subjected to 100 percent burnout by single pulse test, only once, and shall be performed in accordance with Method 4146 of MIL-STD-750 and E = 50 Vdc, minimum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

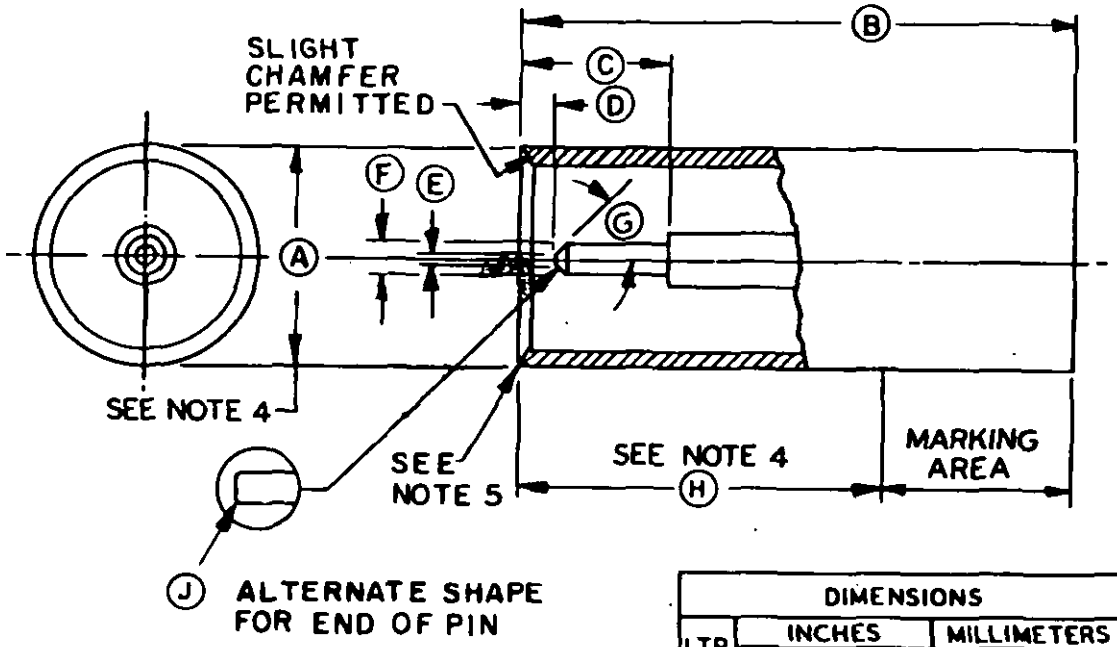
4.2 Test conditions. Unless otherwise specified herein, the test conditions, when applicable, shall be as follows:

P = 1.0 mW ± 5%
f = 24,000 ± 50 mc
Z_m = 500 ± 10 ohms; ± jo ohms
R_L = 100 ± 1 ohms

4.3 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I and II, except the matched pair requirements in subgroup 3 of table I. Qualification testing of either polarity is sufficient to obtain qualification approval of both polarities.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in groups A and B.

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DIMENSIONS				
LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	.215	.220	5.46	5.58
B	.734	.766	18.64	19.45
C	.147	----	3.73	----
D	.011	.028	.28	.71
E	.007	.017	.18	.43
F	.031	.033	.78	.84
G	42°	48°	42°	48°
H	.406	----	10.31	----
J	.007	.017	.18	.43

NOTES:
1. Finish: .0002 in. (.005 mm) tin plate over nickel flash, or .0001 in. (.0025 mm) gold plate or .0001 in. (.0025 mm) silver plate.
2. Axis of center conductor not to deviate from axis of outer conductor referred to its outside diameter more than .004 in. (.101 mm).
3. Standard units shall have the cathode connected to the center conductor. Reversed units shall have the anode connected to the center conductor.
4. Outside diameter, .215 in. (5.46 mm) to .220 in. (5.58 mm) applies for length of dimension H.
5. This edge to be sharp and free from burrs.
6. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

Figure 1. Semiconductor device, diode, types 1N26B, 1N26BR, 1N26BM, and 1N26BMR.

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4.4.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.4.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.5 Methods of examination and test. Methods of examination and test shall be as specified in tables I and II, and as follows:

4.5.1 High-temperature operation. The semiconductor diode shall be placed in the mixer holder. The ambient temperature of the diode, with test conditions specified for the overall noise figure, shall be raised to +150°C min and maintained at this temperature until equilibrium is reached. The NF_O shall then be determined and shall be less than 15 db. The temperature shall then be returned to 25°C ± 3°C at which time NF_O shall be less than 12 db.

4.5.2 Microwave parameters. The L_c, NR_O, and NF_O parameters may be determined by any suitable combination of measured parameters selected from among the following: L_c, NR_O, NF_{if}(actual), and NF_O(actual). A measurement of either L_c or NR_O, but not both, is required. NF_O shall be determined for an assumed or actual intermediate-frequency noise figure (NF_{if}) of 1.5 ± 0.25 db.

4.5.3 Matched pair, "M" suffix (forward polarity). The matched forward pair (M suffix) shall consist of two diodes, tested to requirements of subgroup 3, table I, having the cathode connected to the center conductor.

4.5.4 Matched pair, "MR" suffix (forward and reverse polarity). The matched forward and reverse pair (MR suffix) shall consist of two diodes, tested to requirements of subgroup 3, table I: one diode having the cathode connected to the center conductor (forward polarity) and the second diode having the anode connected to the center conductor (reverse polarity).

Table I. Group A inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Voltage standing wave ratio	4136	Dwg 107-JAN or SCL-5679/1		VSWR	---	1.5	---
Overall noise figure (see 4.5.2)	4126	Test cond. A; Dwg 107-JAN or SCL-5679/1		NF_O	---	11.0	db
Subgroup 3			5				
Conversion loss (see 4.5.2)	4101	Dwg 107-JAN or SCL-5679/1		L_c	---	7.5	db
IF impedance	4116	Dwg 107-JAN or SCL-5679/1		Z(IF)	400	600	ohms
Output noise ratio (see 4.5.2)	4121	Dwg 107-JAN or SCL-5679/1		NR_O	---	2.0	---

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Table I. Group A inspection. - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 3 - Continued							
Matched pair requirements: (See 4.5.3 and 4.5.4)							
Conversion loss unbalance	4101			ΔL_c	---	0.3	db
IF impedance unbalance	4116			$\Delta Z(IF)$	---	25	ohms

Table II. Group B inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1							
Physical dimensions	2066	(See figure 1) Dim. A, C, D and F;	7	---	---	---	---
		Dim. B and H, note 2 (Dim. E, G and J and note 1 are for qualification only)	20	---	---	---	---
Subgroup 2			15				
Thermal shock (temperature cycling)	1051	Test cond. B; T(high) = $+150\ ^{+5}_{-0}$°C		---	---	---	---
Moisture resistance	1021	Omit initial conditioning					
End points: Overall noise figure (see 4.5.2)	4126	Test cond. A; Dwg 107-JAN or SCL-5679/1		NF_o	---	12.0	db
Subgroup 3			15				
Burnout by repetitive pulsing	4141	e_o = 10 v, R_g = 50 ohms; 60,000 pulses min, t_p = 1.0 µsec, PRF optional		---	---	---	---
End points: (Same as for subgroup 2)							
Subgroup 4			15				
Shock	2016	500 G; t ≅ 1 msec 5 blows in each orientation: X_1, Y_1, and Y_2		---	---	---	---
Vibration, variable frequency	2056	15 G; 50 to 2000 cps		---	---	---	---

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Table II. Group B inspection. - Continued.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 4 - Continued							
Constant acceleration	2006	10,000 G; X_1, Y_1, and Y_2 orientations		---	---	---	---
End points: (Same as for subgroup 2)							
Subgroup 5			15				
High temperature operation (see 4.5.1)		$T_A = 150\ ^{+5}_{-0}$°C		---	---	---	---
Subgroup 6			$\lambda = 15$				
High-temperature life (nonoperating)	1031	$T_A = 150\ ^{+5}_{-0}$°C		---	---	---	---
End points: (Same as for subgroup 2)							

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall conform to MIL-S-19500.

6. NOTES

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-S-19500 are applicable to this specification.

Custodians:
Army - EL
Navy - SH
Air Force - 11

Review activities:
Army - EL
Navy - SH
Air Force - 11, 17, 85

User activities:
Army - EL, MU, MI
Navy - WP, MC, CG
Air Force - None

Preparing activity:
Army - EL

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